

Electrical Machines II - Part II

Exhaustive Concept-Wise PYQ Bank

Official University Questions (2022, 2023, 2024, 2025)

Concept 1: Structural Core Geometry & Magnetic Flux

- Q1.** The cross section of the central limb of a shell type transformer is usually designed to have larger cross section to avoid flux saturation. Correct and/or justify. [2024, 2025]
- Q2.** The triplen order flux, sometimes links the tank of transformer and produces loss in the same. Correct and/or justify. [2025]

Concept 2: The Harmonic Dilemma & Core Saturation

- Q3.** With the help of B-H characteristics of magnetic core explain how harmonics are introduced in a transformer. [2023]
- Q4.** Show with a diagram how 3^{rd} harmonics current is produced in a transformer. [2025]
- Q5.** Why a core type three phase transformer is better than other for harmonics related problems? [2022]
- Q6.** How core construction helps in reducing harmonics in a three phase transformer? [2024]
- Q7.** How Delta winding is used to overcome the problems of harmonics in a three phase transformers? [2022]
- Q8.** Show that Delta winding is better in harmonics related issues in a three phase transformer? [2023]
- Q9.** What is the harmonics related problem with Yy connected transformer without neutral? Explain with proper diagram. [2022]
- Q10.** The $9n^{th}$ harmonics of magnetic flux are always in phase though its $7n^{th}$ harmonics have a reversed phase sequence compared to fundamental. Correct and/or justify. [2024]

Concept 3: Spatial Vector Topology & Vector Groups

- Q11.** Explain the importance of vector groups for three phase transformers. [2023]
- Q12.** Vector group formation is unavoidable with 30 degree phase shift in case of Delta-star or Star-delta transformer. Correct and/or justify. [2024, 2025]
- Q13.** Delta-star is more preferable than star-star without neutral three phase transformer. Correct and/or justify. [2024, 2025]
- Q14.** Draw the connection diagram and phasor diagram of the following connections:
- Yy6, Dz6, and Dy1 [2022]
 - Dd6, Dz0, and Dy11 [2023]
 - Dy1, Yz1, and Yd1 [2024]

Concept 4: Parallel Operation Constraints

- Q15.** What will happen if a Dy1 connected transformer is paralleled with a Yd11 connected transformer? Show how they can be connected properly with connection modification. [2022, 2025]
- Q16.** What will happen if a Dy1 connected transformer is paralleled with a Dy11 connected transformer? Show how they can be connected properly with connection modification. [2023]
- Q17.** Yy0 and Dd6 connected transformers cannot be connected in parallel - explain. [2024]

Concept 5: Special Phase Conversions (Scott Connection)

- Q18.** If two identical single-phase transformers have tapings at 50% and 86.7% then show that they can be connected in Scott connection to convert a three to a two phase system. [2022]
- Q19.** Is it possible to connect two single phase transformers, with one having a center tap and the other with no tap, in a Scott-connected transformer? Discuss with proper diagrams. [2023]
- Q20.** Show that by using two single phase transformers with proper taps, three phase power supply is possible from a two phase supply. [2024]
- Q21. Numerical Problem:** In Scott-connected transformers, teaser transformer supplies 0.75 leading power factor load of 30 kW at 220 V and main transformer supplies 0.85 power factor lagging load of 50 kW at 220 V, from a three phase input line voltage of 3300V. Determine the input 3-phase line currents. Neglect magnetizing currents and the leakage impedance drops. Draw input current phasors computed for above. [2022]
- Q22. Numerical Problem:** In Scott-connected transformers, teaser transformer supplies 0.75 lagging power factor load of 20 kW at 220 V and main transformer supplies 0.85 power factor leading load of 40 kW at 220 V, from a three phase input line voltage of 3300V. Determine the input 3-phase line currents. Neglect magnetizing currents and the leakage impedance drops. Draw input current phasors computed for above along with a Scott-connection diagram. [2023]
- Q23. Numerical Problem:** In Scott-connected transformers, teaser transformer supplies 0.75 leading power factor load of 25 kW at 220 V and main transformer supplies 0.75 power factor lagging load of 45 kW at 220 V, from a three phase input line voltage of 3300V. Determine the input 3-phase line currents. Neglect magnetizing currents and the leakage impedance drops. Draw input current phasors computed for above along with a Scott-connection diagram. [2024]
- Q24. Numerical Problem:** In Scott-connected transformers, teaser transformer supplies 0.75 leading power factor load of 25 kW at 220 V and main transformer supplies 0.75 power factor lagging load of 45 kW at 220 V, from a three phase input line voltage of 6600V. Determine the input 3-phase line currents. Draw input current phasors computed for above along with a Scott-connection diagram. Neglect magnetizing currents and the leakage impedance drops. [2025]

Concept 6: Dynamic Voltage Regulation (OLTC & Tappings)

- Q25.** Discuss the placement of taps in a three phase transformer. [2023]
- Q26.** Why high voltage winding of a transformer is preferred for tappings of tap-changer? (Also asked as: Taps should be provided at the middle of the high voltage winding - explain). [2022, 2024, 2025]
- Q27.** Component wise what is the basic difference of an Off-load and an ON-load tap changer? / Write the advantages and disadvantages of using off-load and on-load type tap-changers. [2023, 2024]
- Q28.** Why either resistor or reactor is used in an On-load tap changer(OLTC)? [2022]
- Q29.** Describe the operation of a reactor type tap changer. [2023, 2025]
- Q30.** Describe the operation of a resistor type tap changer (also asked as: Draw and explain two resistor type on-load tap changer schemes). [2022, 2024]
- Q31.** Write the merits and demerits of resistor-type on-load type tap-changers. [2025]

Concept 7: High-Frequency Transients, Testing & Anchors

- Q32.** Development of voltage stress along the winding of a three phase transformer for input impulse and RMS voltage. And also write the measure to be taken to withstand it. [2022, 2023, 2024]
- Q33.** Full wave and chopped wave impulse voltage tests on a transformer for detection of faults in the winding. [2022, 2023, 2025]
- Q34.** What is oscillating neutral problem? - Explain with proper diagram. (Also asked as: Neutral oscillation in 3-phase transformers). [2023, 2025]
- Q35.** Write short notes on: Tertiary windings and its applications. [2024, 2025]